

STANDARD OPERATING PROCEDURE

Furnace FRN-402 — Reformer Unit Startup Sequence

Document ID	SOP-001	Plant / Unit	BHLV — Reformer Unit, Train A
Document Type	Standard Operating Procedure	Date Issued	12 March 2024
Revision	Rev 1.0	Classification	INTERNAL — CONTROLLED

1. Purpose & Scope

This procedure defines the controlled startup sequence for Gas-Fired Reformer Furnace FRN-402 at the Bharat Heavy Industries Ltd. Visakhapatnam Refinery & Chemical Complex (BHLV). It applies to all cold-start, warm-start, and post-maintenance restart operations. The procedure must be followed in strict sequence. No step may be bypassed without written dispensation from the Plant Manager (R. Krishnan) and the Safety & Loss Prevention Officer (S. Mehta).

2. Referenced Equipment

Tag	Equipment	Location	Design Pressure
FRN-402	Gas-Fired Reformer Furnace	Reformer Unit, Bay 4	—
HE-101	Shell & Tube Heat Exchanger — Crude Preheat Train	Preheat Section	42.0
PMP-203	Centrifugal Pump — HFO Feed Line	Pump House B	18.5
V-301	Vertical Pressure Vessel — Propane Surge Drum	Tank Farm West	24.0
VLV-507	Control Valve — Feed Header	Reformer Inlet Header	—

3. Pre-Startup Checks

All items below must be confirmed and initialled by the Field Technician (T. Das) before proceeding to Section 4:

- 3.1 Confirm HE-101 outlet temperature is within operating range: 160–175 °C. Record reading from DCS tag TIC-101A. If reading is outside range, notify K. Rajan immediately.
- 3.2 Confirm PMP-203 discharge pressure is stable at 17.8–18.5 bar g. Check mechanical seal — no visible weep permitted. Record pump amps.
- 3.3 Verify V-301 level gauge reads 35–65% on LIC-301. Confirm pressure at PIC-301 is 21.5–24.0 bar g. Pressure outside this band must be reported to S. Mehta before startup.

- 3.4** Confirm VLV-507 is in AUTO mode and feed header pressure is stable.
- 3.5** Confirm all maintenance work orders on FRN-402 and associated equipment are closed and signed off by A. Nair. No open hot-work permits within 15 m of FRN-402.
- 3.6** Conduct gas detector sweep of furnace enclosure and adjacent areas. No ignition source permitted if gas reading exceeds 10% LEL.

4. Startup Sequence

- Step 1:** Open furnace purge valve. Purge FRN-402 enclosure with steam for minimum 15 minutes. Confirm purge completion with combustion analyser ($O_2 < 0.5\%$). Log time: _____
- Step 2:** Enable pilot gas supply. Confirm pilot flame established on all 8 burners via flame scanner indication on DCS. Time to flame: _____ (maximum 90 seconds per burner; abort and re-purge if exceeded).
- Step 3:** Ramp fuel gas flow via FIC-402 in 10% increments at 5-minute intervals until duty setpoint reached. Monitor firebox temperature TI-402A/B/C continuously during ramp.
- Step 4:** Confirm feed flow through HE-101 is stable. Cross-check DCS tag FIC-101 against local flow indicator. Deviation $> 5\%$ requires investigation before continuing ramp.
- Step 5:** Transfer FRN-402 to AUTO control once all process parameters are within normal operating range. Notify R. Krishnan and log startup completion time in shift handover record.

5. Abnormal Conditions

- CRITICAL:** If gas detector alarm activates during startup at any stage, immediately initiate emergency shutdown (ESD) using panel button ESD-FRN-402. Do not attempt manual restart without investigation sign-off by S. Mehta and K. Rajan. Refer to NMR-2024-012 for details of the March 2024 gas accumulation incident near V-301 that preceded an unplanned furnace trip — conditions described there may recur under similar atmospheric and pressure conditions.
- If V-301 pressure exceeds 24.0 bar g during startup, reduce PMP-203 flow rate by 20% and notify A. Nair. Do not override PIC-301 high-pressure interlock. Refer INS-2024-061 for V-301 current inspection status.

6. Authorisation

Role	Name	Signature	Date
Senior Process Engineer	K. Rajan	_____	12 Mar 2024
Safety & Loss Prevention	S. Mehta	_____	12 Mar 2024
Maintenance Supervisor	A. Nair	_____	12 Mar 2024
Plant Manager	R. Krishnan	_____	14 Mar 2024

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